

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.



Dataset at a Glance

3,900

Purchases

Demographics

Age, Gender, Location, Subscription

18

Features

Purchase Details

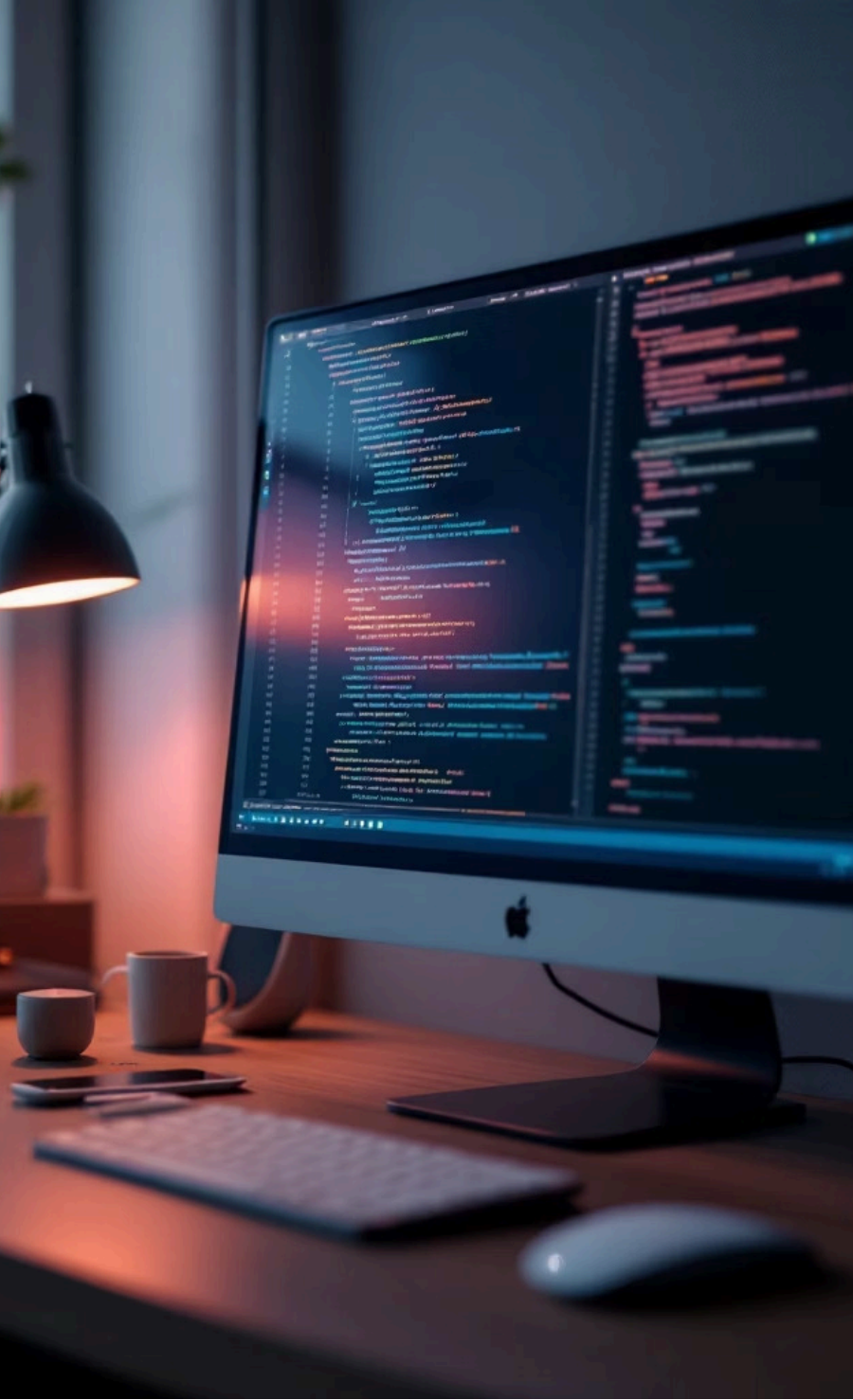
Item, Category, Amount, Season,
Size, Color

37

Missing Ratings

Behavior

Discounts, Frequency, Ratings,
Shipping



Data Preparation in Python

01

Load & Explore

Imported with pandas; used `df.info()` and `.describe()`

02

Clean & Impute

Missing Review Ratings filled with **median per category**

03

Feature Engineering

Added `age_group` and `purchase_frequency_days` columns

04

Export to PostgreSQL

Cleaned DataFrame loaded into database for SQL analysis

SQL Analysis: Revenue Insights



Revenue by Gender

Compared total revenue from male vs. female customers.

Shipping Type Comparison

Standard vs. Express shipping — average purchase amounts compared.

SQL Analysis: Customer Segments



New

First-time buyers



Returning

Moderate purchase history



Loyal

>5 purchases; more likely to
subscribe

Subscribers vs. non-subscribers also compared on **average spend** and **total revenue**.

SQL Analysis: Products & Discounts

Top 5 by Rating

Products with highest average review scores identified.

Top 3 per Category

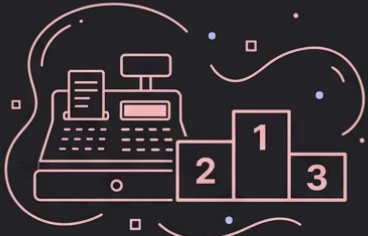
Most purchased products listed within each category.

Discount-Dependent Products

5 products with the **highest % of discounted purchases** flagged.



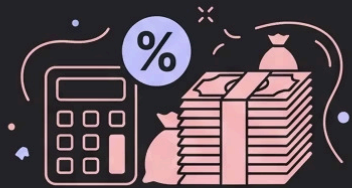
Top Rated Products
Highest avg review ratings



Best Sellers per Category
Top 3 per category



Top Rated Products
Highest avg review ratings



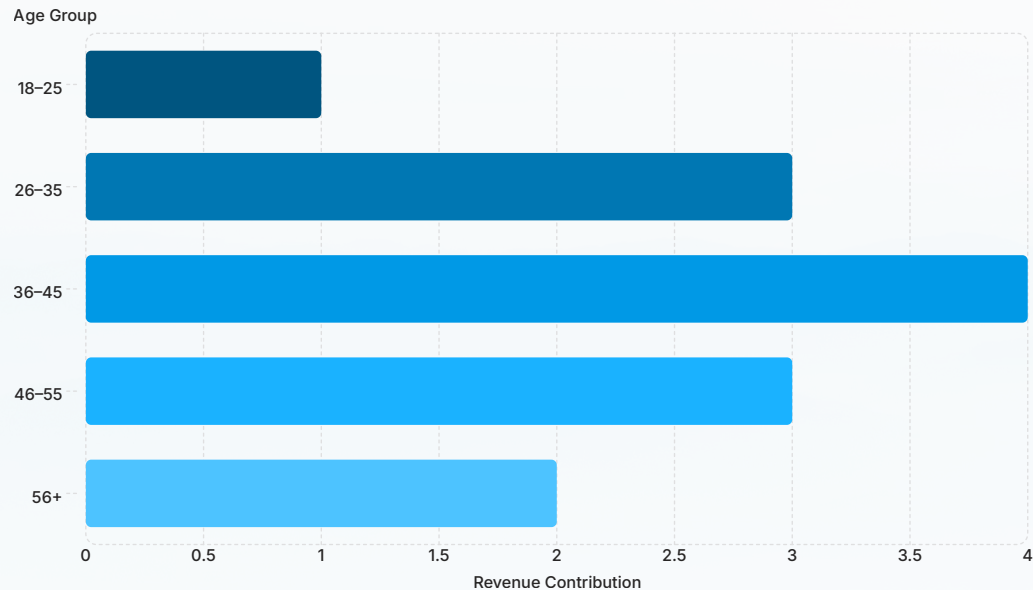
Discount Reliant Products
Highest % of discounted sales

SQL: High-Spending Discount Users

Customers who used discounts but **still spent above the average purchase amount** — a key segment for targeted upsell strategies.



Revenue by Age Group



Age-Driven Revenue

Total revenue contribution calculated per age group — revealing which demographics drive the most value.

- Focus marketing on **high-revenue age groups** for maximum ROI.

Power BI Dashboard

All findings consolidated into an interactive dashboard for real-time business monitoring and decision-making.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → move to "Loyal" segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users